Suleyman Hajizadeh

COMPUTATIONAL BIOLOGIST & BIOINFORMATICIAN

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RESEARCH INTERESTS

Computational genomics & transcriptomics · Structural bioinformatics & computational biophysics · Statistical machine learning applied to cancer biology · Reproducible pipeline engineering · Molecular evolution & phylogenomics

EDUCATION

Western Caspian University
Baku, Azerbaijan · 2022 – Jun 2026

Bachelor of Science in Biology

- Relevant coursework: Molecular Biology, Genetics, Biochemistry, Biostatistics, Computational Methods in Biology
- Independent research in computational cancer genomics conducted in parallel with degree coursework

RESEARCH & TECHNICAL PROJECTS

High-Throughput RNA-Seq Pipeline — Triple-Negative Breast Cancer
2024 – Present

Institute of Molecular Biology & Biotechnology (IMBB), Baku

- Developed a **Snakemake** workflow (SRA → FastQC → Trimmomatic → HISAT2/GRCh38 → featureCounts → DESeq2) for fully reproducible bulk RNA-seq analysis of GSE183947 (TNBC cohort)
- Implemented **DESeq2** differential expression with statistical commentary: derived negative binomial model assumptions, BH-FDR correction mathematics, and VST vs. rlog trade-offs in a dedicated *Statistical Methods* primer
- Designed a nested Leave-One-Out Cross-Validation classifier with feature selection inside training folds to prevent data leakage; permutation test empirical p-value < 0.05

Computational Phylogenomics CLI — COL1A1 Vertebrate Evolution
2025

- Built `phylo_pipeline.py` — a **pure Python** command-line tool that fetches CDS sequences from NCBI Entrez, performs pairwise alignment, and computes **K2P and JC69 distance matrices from mathematical formulas**
- Implemented the **Neighbor-Joining algorithm from scratch** in NumPy; reconstructed the evolutionary history of COL1A1 across 7 vertebrate species (*H. sapiens, Pan troglodytes, M. musculus, R. norvegicus, B. taurus, G. gallus, D. rerio*)
- Outputs: Newick tree file (`.nwk`) and high-resolution cladogram PNG — **no MEGA GUI dependency**

Protein Computational Biophysics Toolkit — 2025

- Parsed crystallographic PDB coordinate files with Biopython; extracted Cα atoms and computed 3D Euclidean distance matrices and backbone RMSD between structural domains
- Implemented **hydrophobic core detection** using Kyte-Doolittle scale weighted by spatial packing density (8 Å threshold)
- Coded a **Miyazawa-Jernigan statistical contact potential** loop to estimate non-covalent folding free energies at residue-pair resolution
- Computed Pearson correlation between Cα packing density and crystallographic B-factors ($r = -0.37$ *for insulin 1COH*), mathematically validating core rigidity

AlphaFold2 Structural Analysis — AKT1 Kinase (TNBC) — 2024

- Predicted 3D structure of AKT1 hub kinase using ColabFold v1.6.1 (AlphaFold2 multimer); mean pLDDT = 65.61 across full chain
- Generated per-residue pLDDT confidence plots, Ramachandran backbone torsion analysis, Cα contact maps, and PyMOL publication-quality renders

TECHNICAL SKILLS

Languages	Python (Advanced) · R (Intermediate) · Bash/Shell scripting
Sci. Python	Biopython · NumPy · Pandas · SciPy · Scikit-Learn · Matplotlib
R / Bioconductor	DESeq2 · WGCNA · edgeR · pROC · ggplot2 · limma
Pipelines	Snakemake · Conda · Git/GitHub · SRA Toolkit · HISAT2 · featureCounts
Structural Bio	AlphaFold2 (ColabFold) · PyMOL · PDB parsing · RMSD · contact maps
Statistics / ML	Negative Binomial model · BH-FDR · Nested CV · Random Forest · PCA · GSEA
From Scratch	K2P distance matrix · JC69 model · NJ algorithm · MJ contact potential · hydrophobic core detection

AFFILIATIONS & TRAINING

Institute of Molecular Biology & Biotechnology (IMBB) — Baku, Azerbaijan · 2023 – Present

Bioinformatician (independent researcher)

Data Analysis for Life Sciences — Harvard University · edX · Expected Jul 2026

Bioinformatics MicroMasters — University of Maryland Global Campus · edX · Expected Aug 2026

LANGUAGES

Azerbaijani	English	Russian
Native	Professional (IELTS target 7.5+, Jul 2026)	Conversational